(iv)It combines with acidic oxides such as $SiO_2$ and $P_4O_{10}$ to form $CaSiO_3$ and $Ca_3(PO_4)_2$, respectively.

$CaO + SiO_2 \rightarrow CaSiO_3$

$6CaO + P_4O_{10} \rightarrow 2Ca_3(PO_4)_2$

### Uses

Calcium oxide is used

(i) to manufacture cement, mortar and glass.

(ii) in the manufacture of sodium carbonate and slaked lime.

(iii) in the purification of sugar.

(iv) as a drying agent.

## 5.6.2 Calcium hydroxide

### Preparation

Calcium hydroxide is prepared by adding water to quick lime, CaO.

### Properties

It is a white powder. It is sparingly soluble in water. The aqueous solution is known as lime water and a suspension of slaked lime in water is known as milk of lime.

When carbon dioxide is passed through lime water, it turns milky due to the formation of calcium carbonate.

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

On passing excess of carbon dioxide, the precipitate dissolves to form calcium hydrogen carbonate.

$CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$

Milk of lime reacts with chlorine to form hypochlorite, a constituent of bleaching powder.

$2Ca(OH)_2 + 2Cl_2 \rightarrow$

$CaCl_2 + Ca(OCl)_2 + 2H_2O$

### Uses:

Calcium hydroxide is used

(i) in the preparation of mortar, a building material.

ii) in white wash due to its disinfectant nature.

(iii) in glass making, in tanning industry, in the preparation of bleaching powder and for the purification of sugar.

## 5.6.3 Gypsum ($CaSO_4.2H_2O$)

Gypsum beds were formed due to the evaporation of water from the massive prehistoric sea basins. When water evaporates, the minerals present in it become concentrated, and crystallise.



**Figure 5.11 A Gypsum Quarry**

### Properties of Gypsum

- Gypsum is a soft mineral, which is moderately soluble in water. The solubility of this mineral in water is



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